FloatSat

Welcome

To the satellite builders group

Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

Informationstechnik für Luft- und Raumfahrt
Aerospace Information Technology

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Who is who in Info-8

Who is who in Info-8

Sekretary and Administration:
Anna Gonel

Please Register in our University

1. We need it

2. You will get access to online courses
   https://wuecampus2.uni-wuerzburg.de/moodle/
   (Shall we search for you??)

3. You will get access to online library and many e-books
   http://katalog.bibliothek.uni-wuerzburg.de

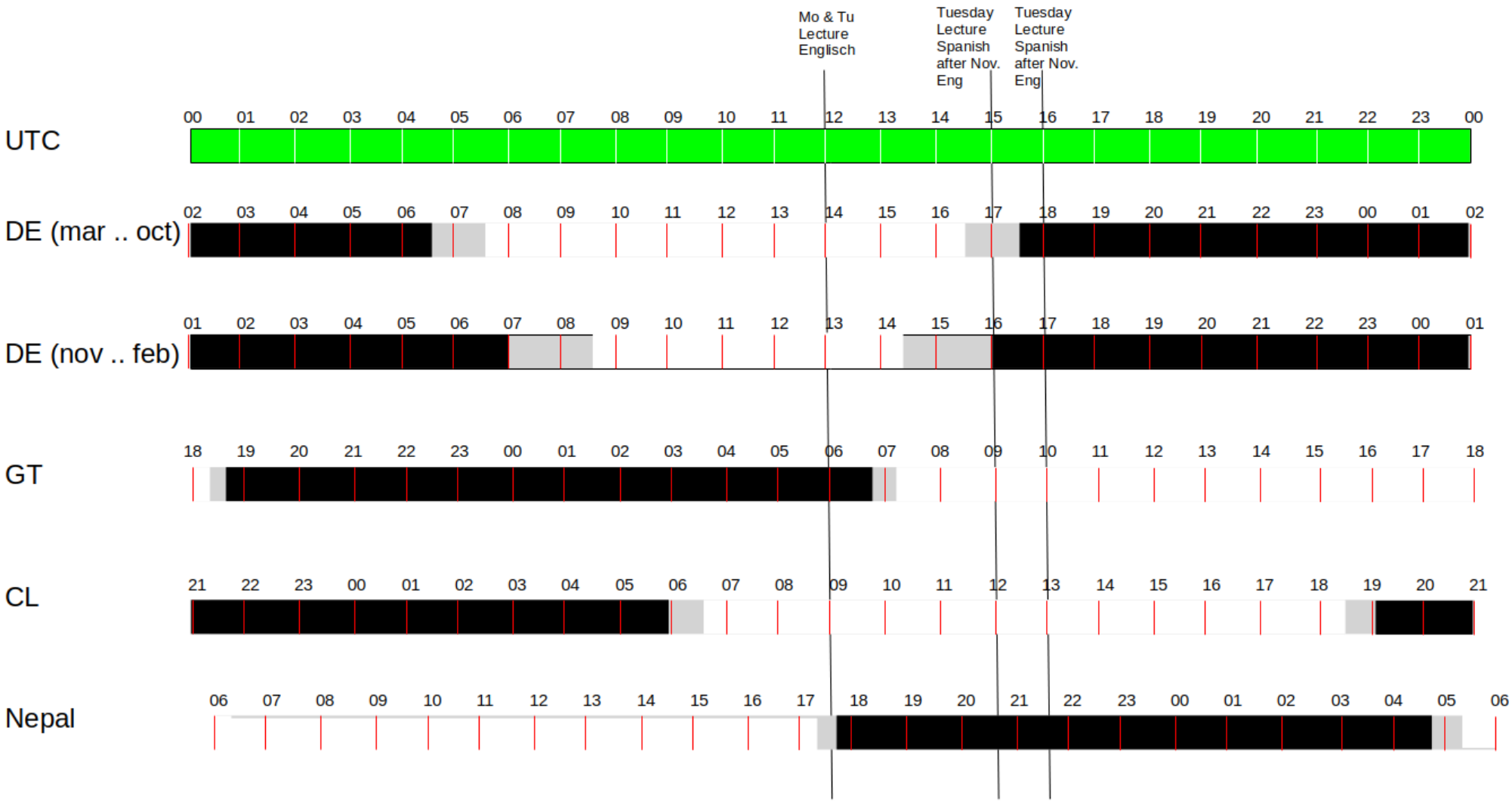
4. You will get an account with email, cloud-services, etc etc

5. You will get a Certificate from our university

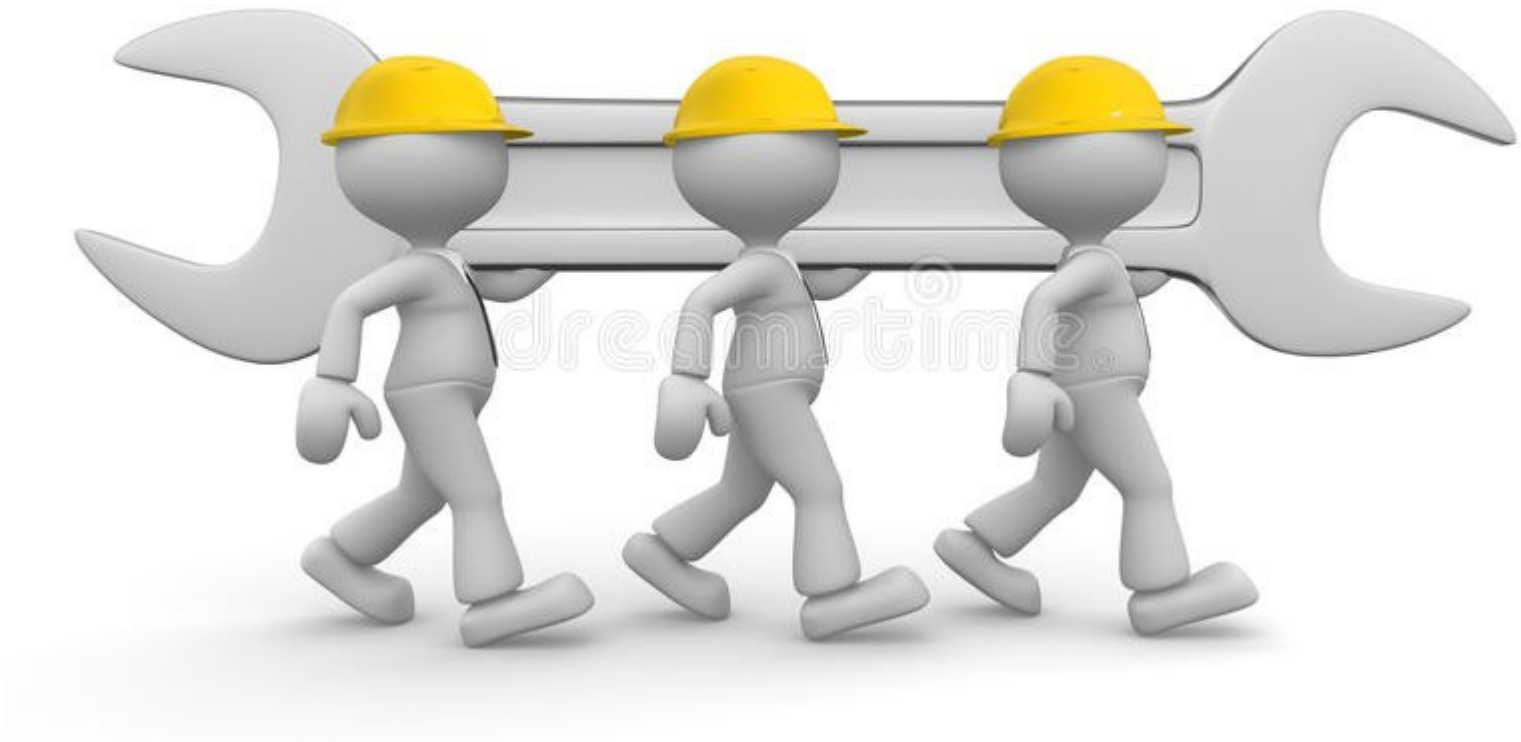
6. you will get VPN and Eduroam

6. You will get a bus-card :) German Insurance :)

Lecture Times



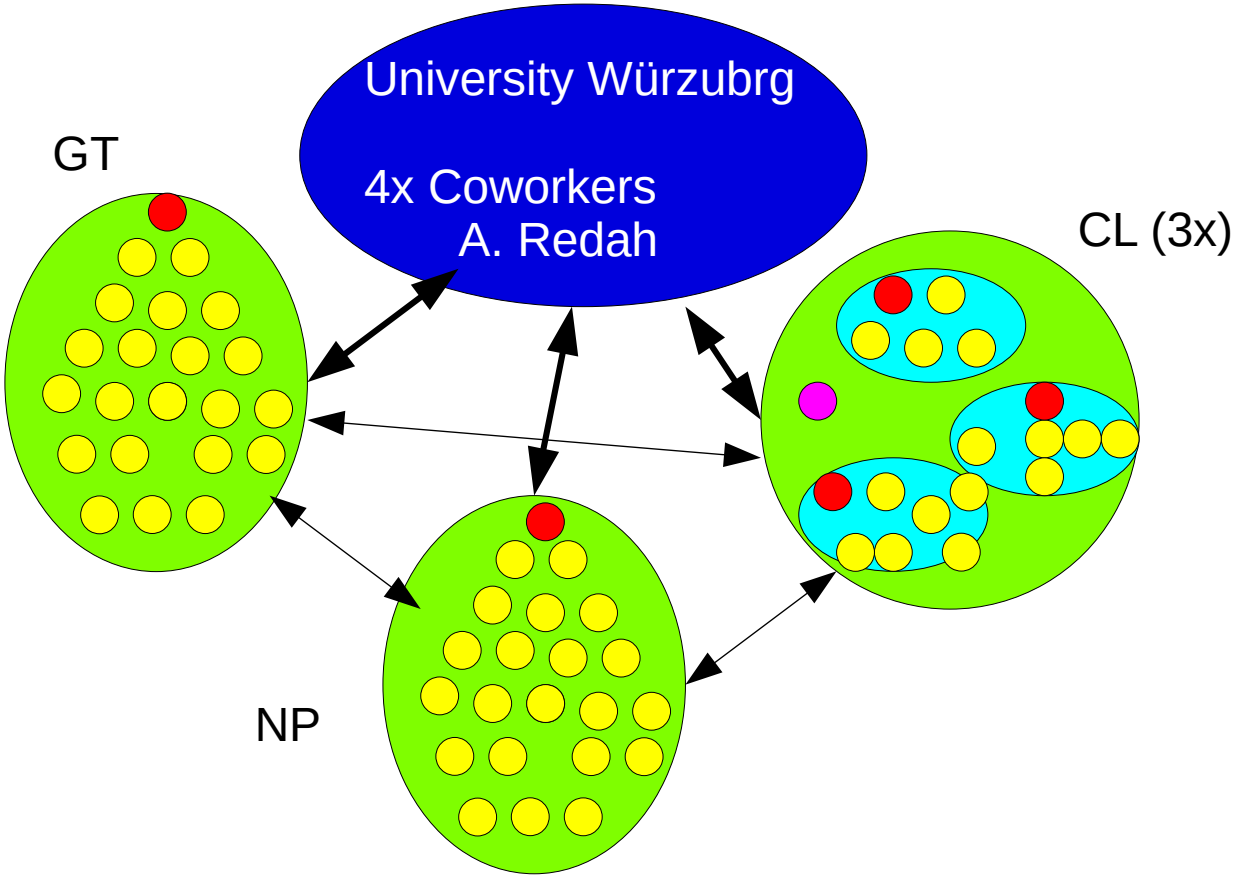
Teamwork



Coordination effort = F(n^2, Distance, complexity, interdisciplinary)
in our case: Terrible!
→ divide

Teamwork

Disciplines:
- Software
  - Embedded
  - GUI
  - Comms
- Electronics
  - Hardware
  - Analog
  - Power
- Mechanical
- Control
- Physics
- Mathematics
- Management



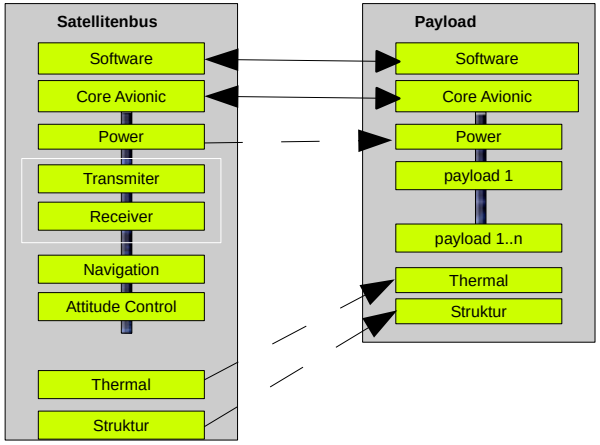
Our Tasks

Mission components

Mission

Space Segment

Platform / Bus



Payload

Launch Segment

Deployment container

Launcher

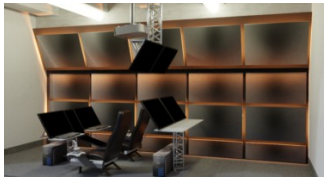
Ground Segment

Ground Station

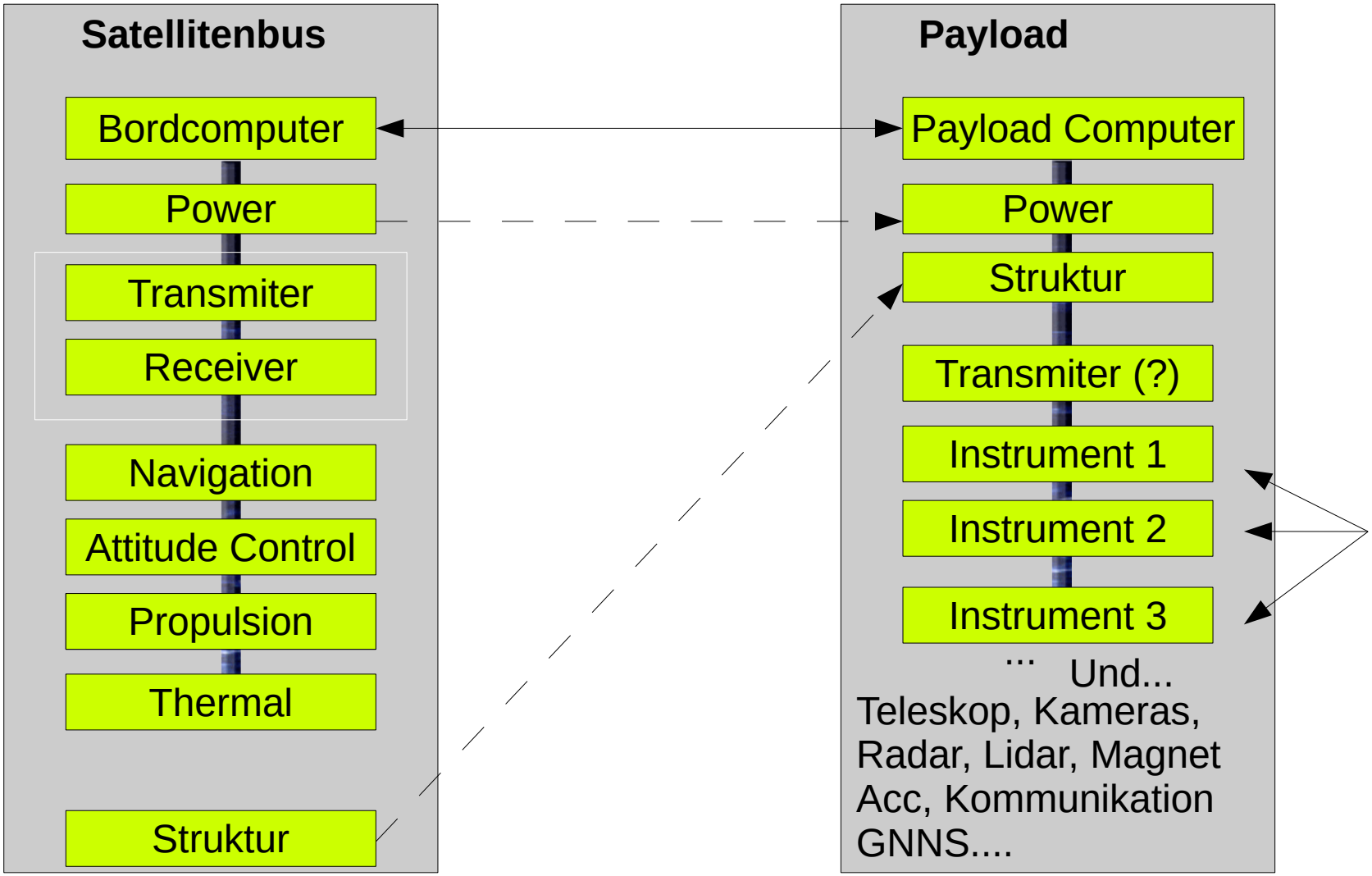
Control Software

- Orbit predictor
- Planer
- Antenna Cntr.
- Enc/Decoder
- Data Router
- Data Bank
- TM Interface
- TC Interface
- Payload Intf.

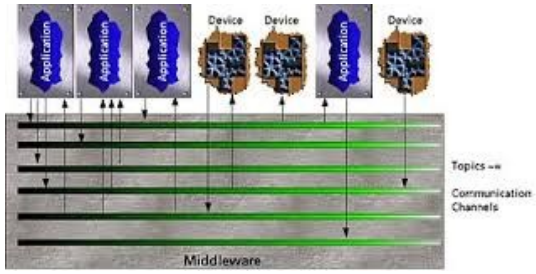
Control Room



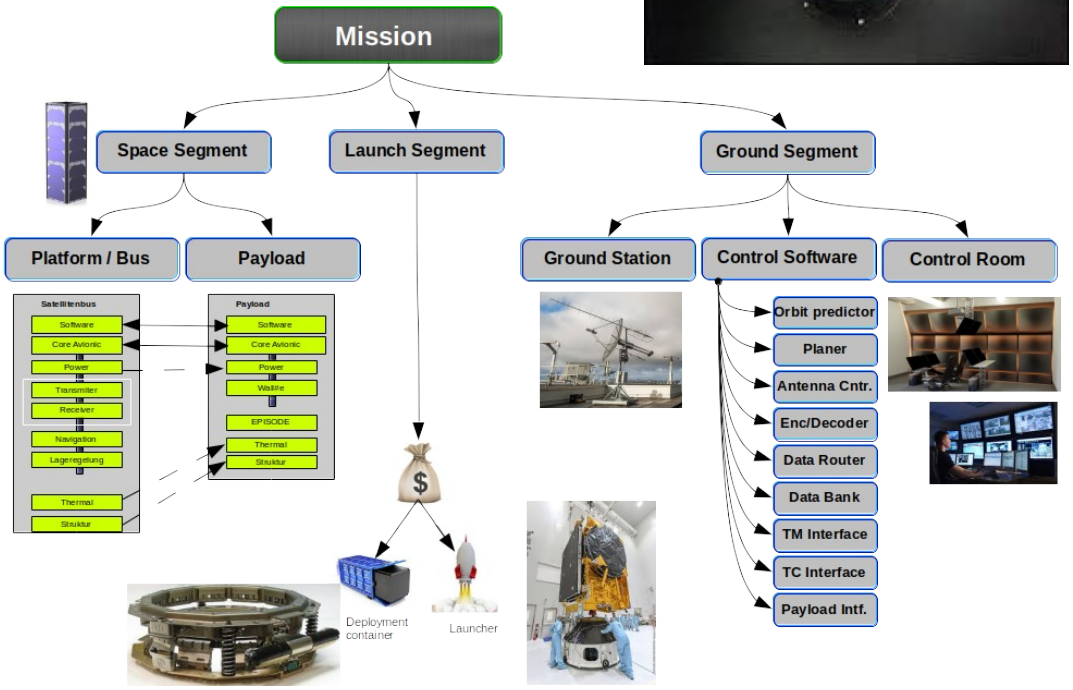
Space segment



Our contribution



- Materials
- Software
- Lectures
  - Ground Segment
  - Space Segment
  - (Launch Segment)
- Exercises Support



Your contribution

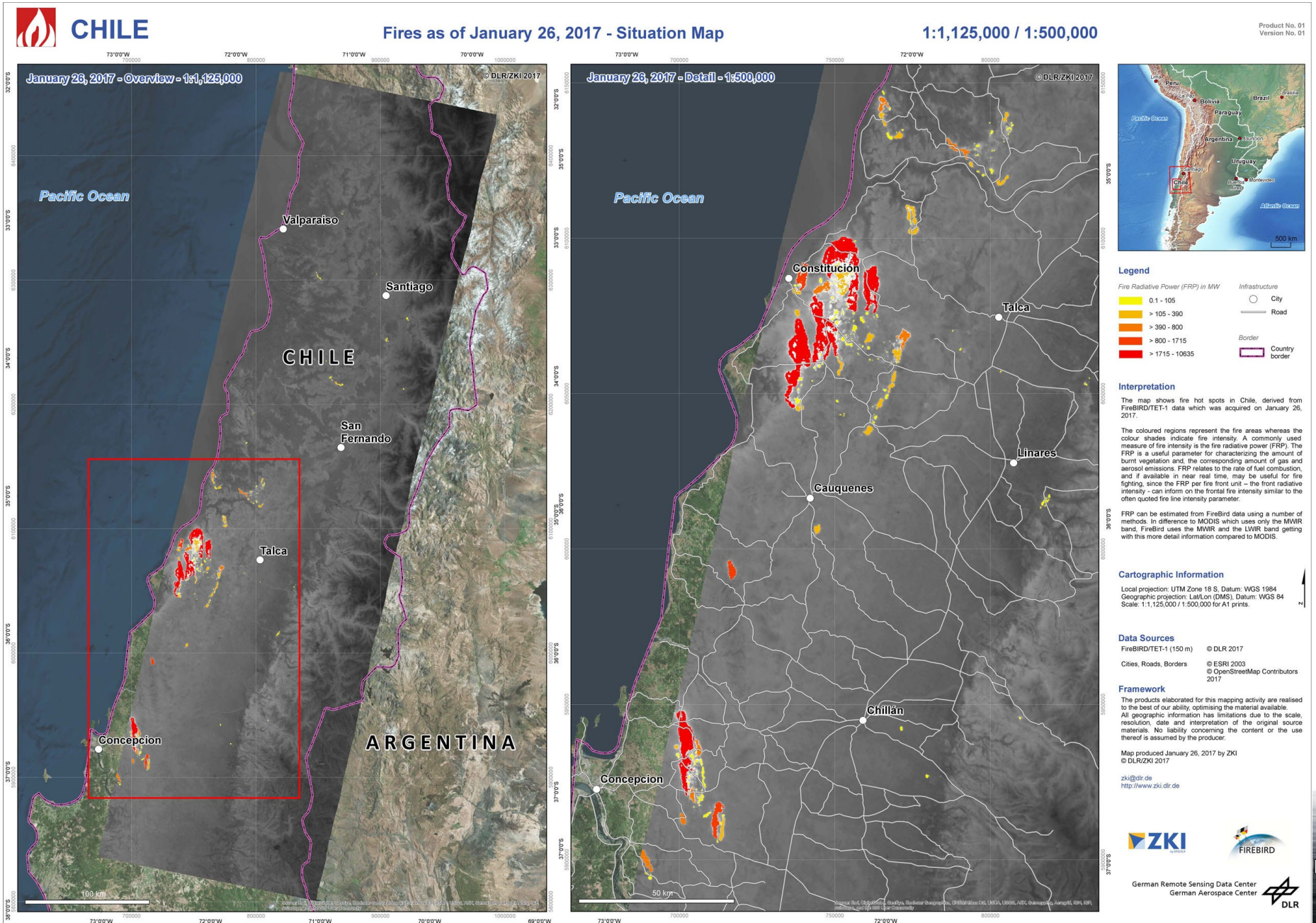
Disciplines:
- Software
  - Embedded
  - GUI
  - Comms
- Electronics
  - Hardware
  - Analog
  - Power
- Mechanical
- Control
- Physics
- Mathematics
- Management
  - Coordination
  - Administration
  - Controlling
  - Finances
  - Contracts

- Embedded Programming
- GUI & Ground Station Programming
- Electronics
- Structure & Mechanism
- Control & Steering
- Physical and Mathematical Models
- Management and Coordination

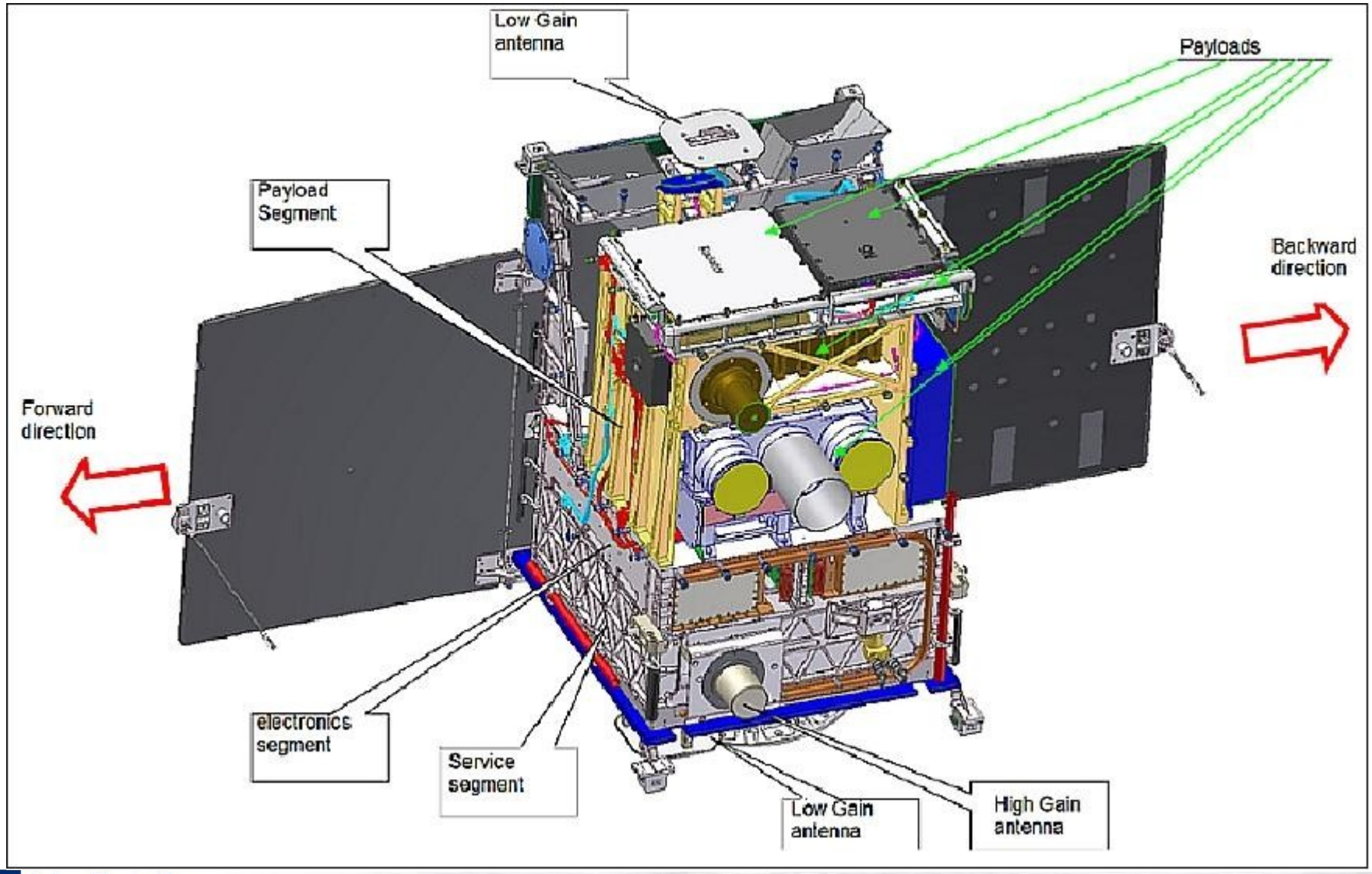
- Else:
  Tel us what you can do,
  we/your local coordinator
  will „use“ it

Space Segment

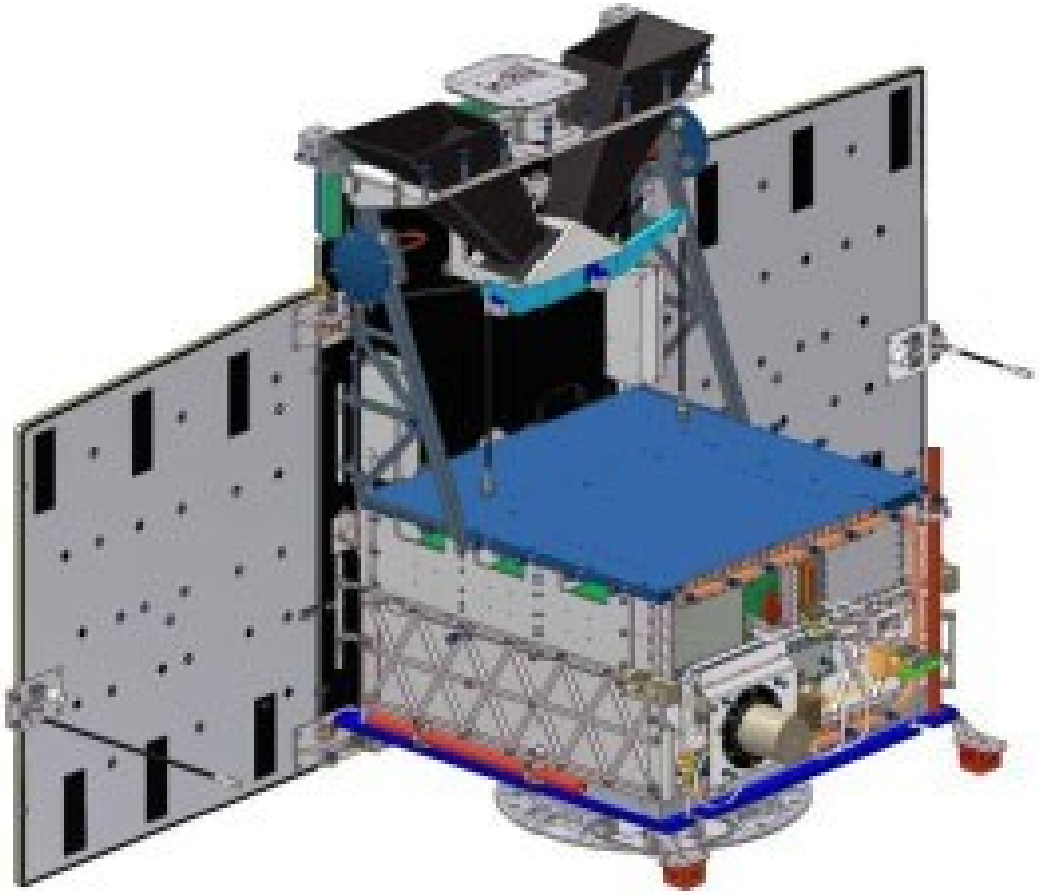
Mission: Find fire

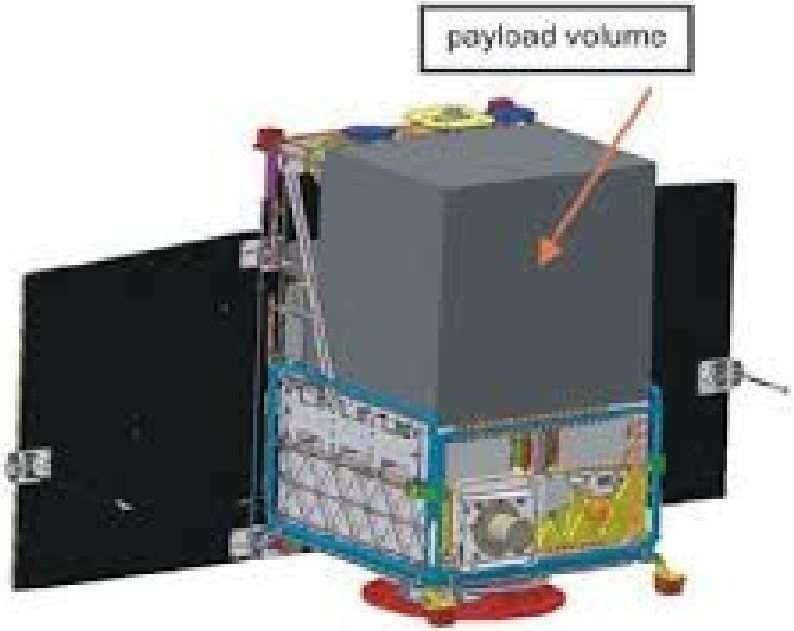


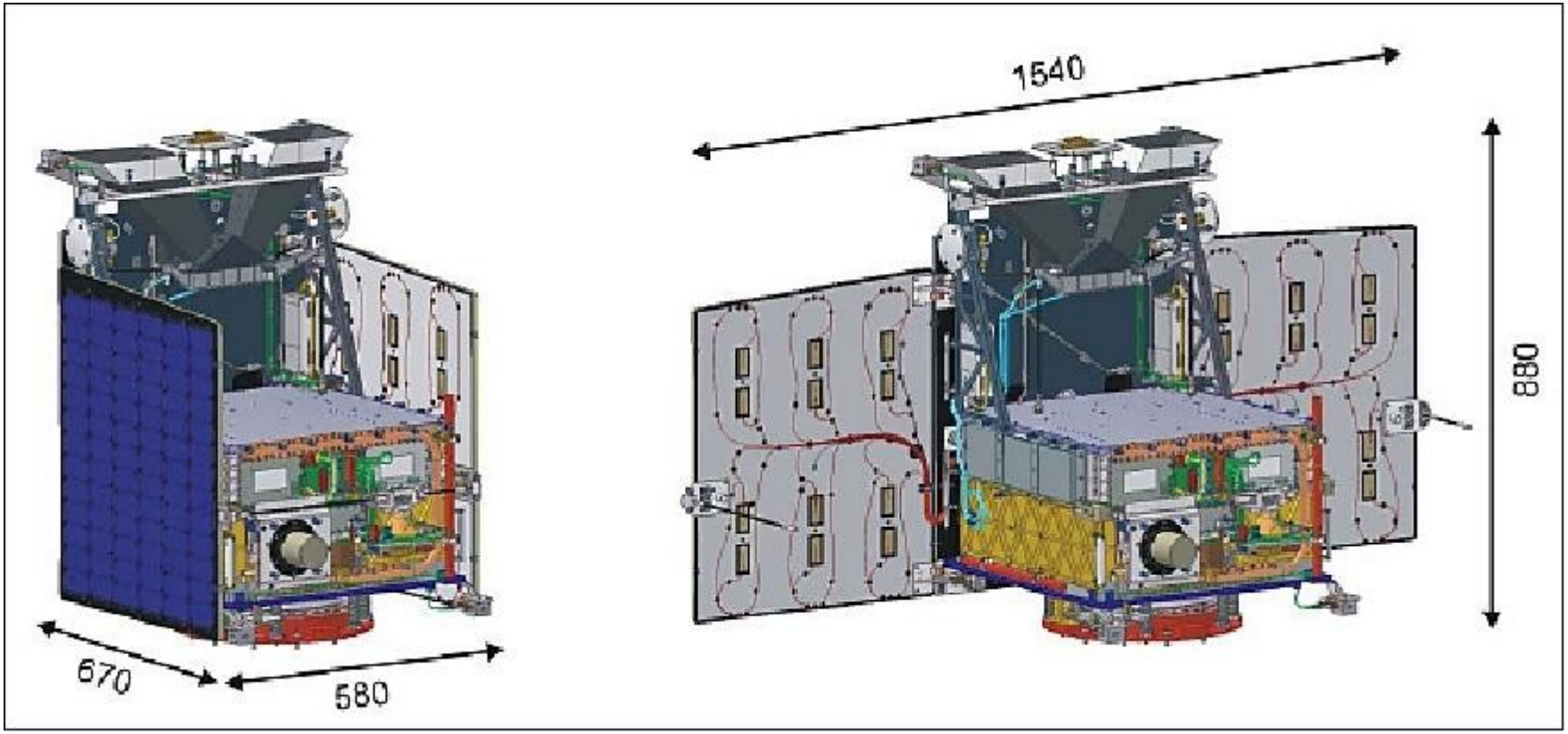
Space Segment



Platform / Bus

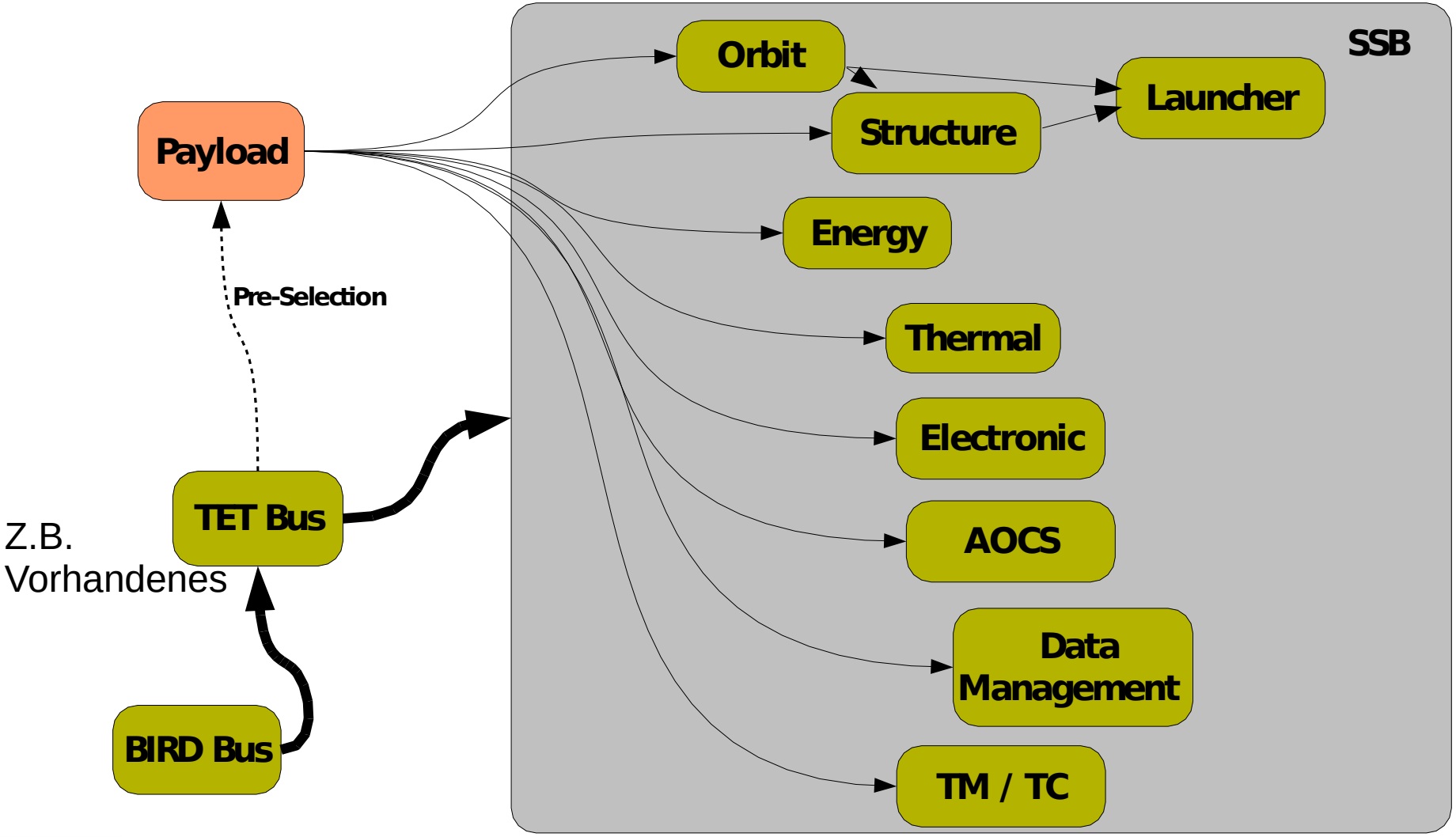




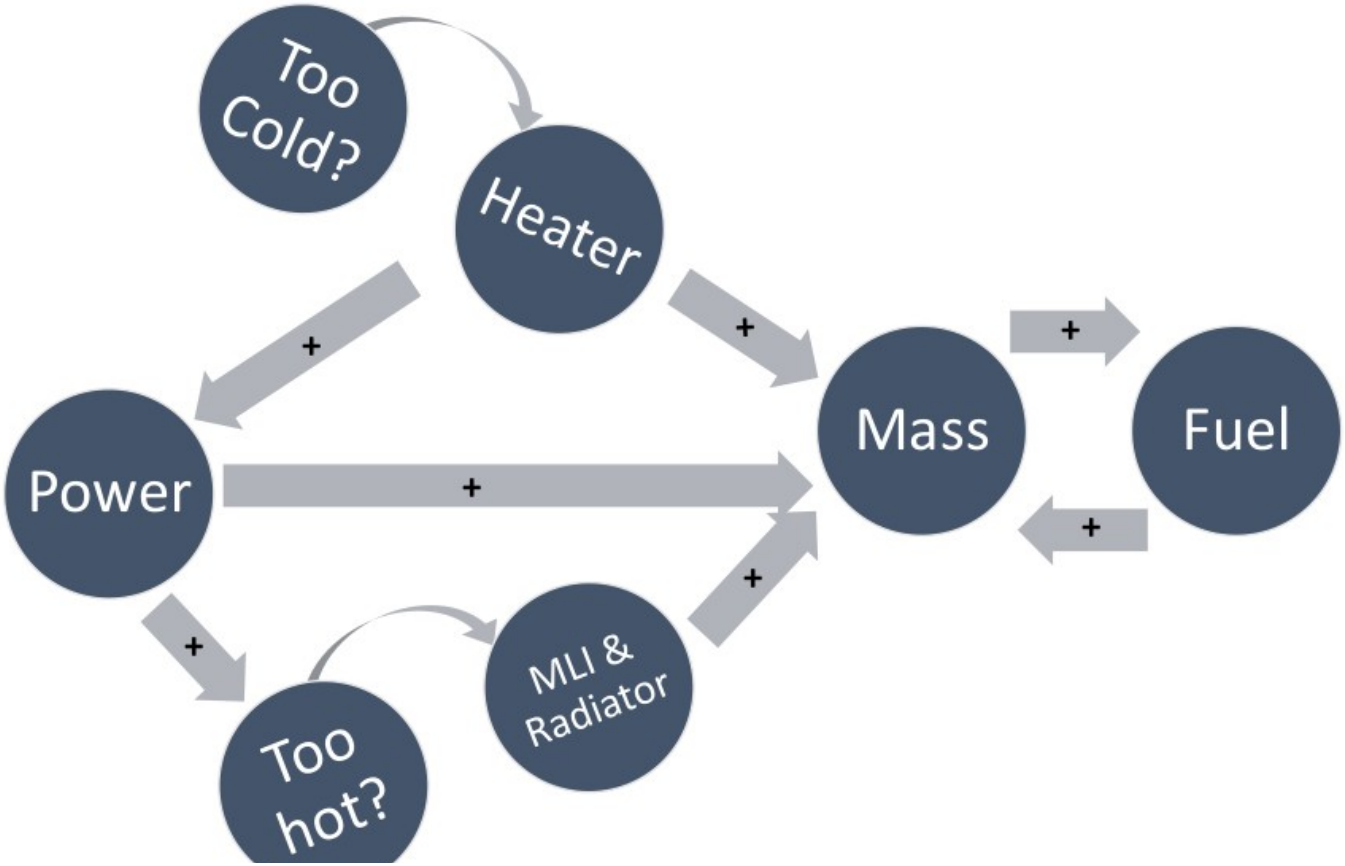

1540
880
670
580

Space Segment

Bus – Payload Dependabilites



1. The problem of interactions



2. Two types of engineering

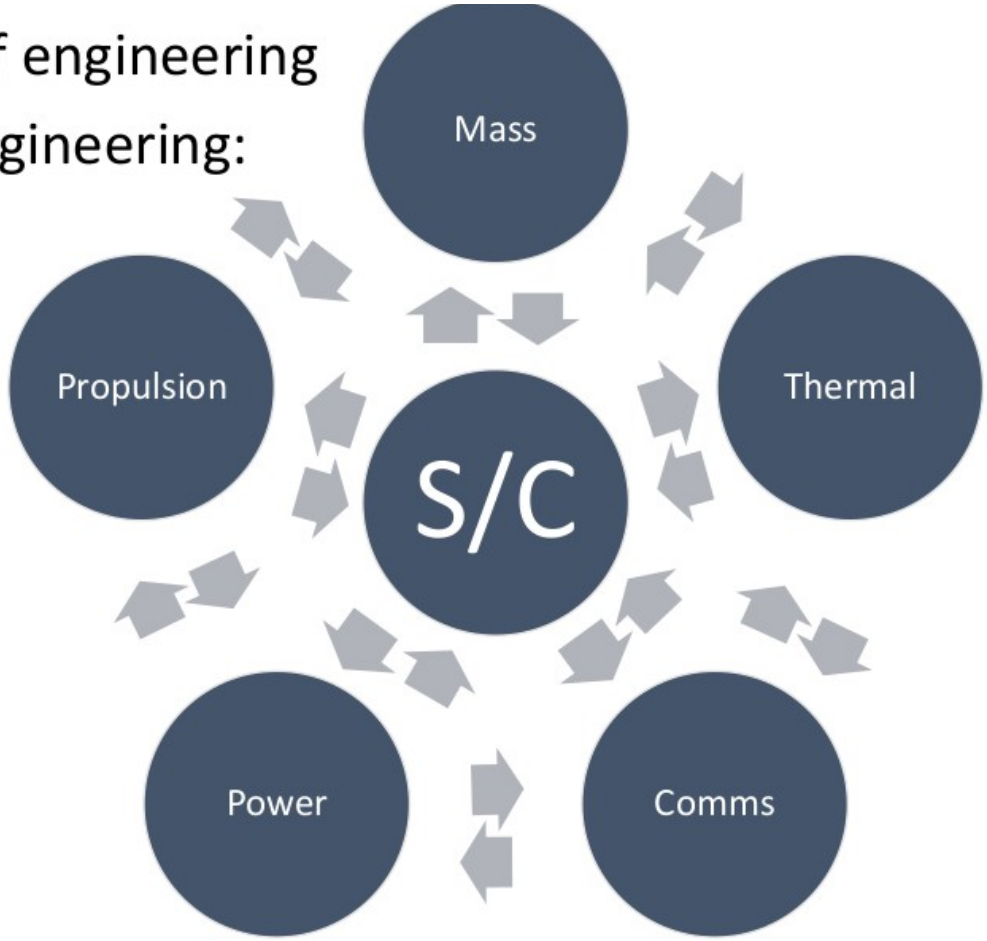
- Linear engineering:

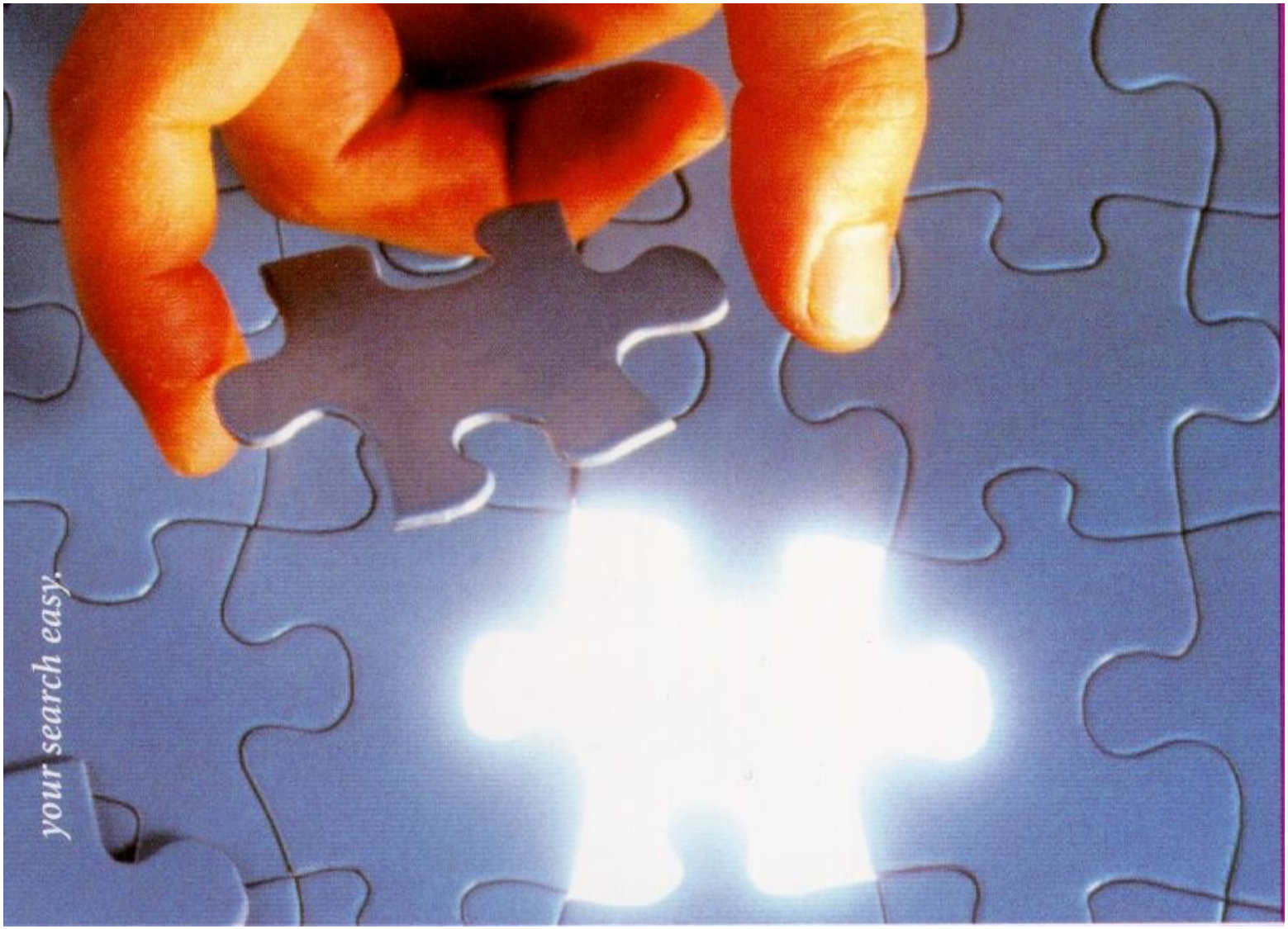
Traditionnal way of designing

- Easy to do
- Inefficient in Space

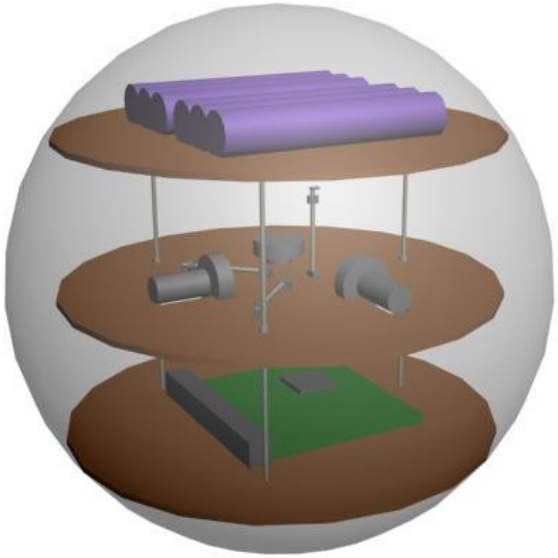
2. Two types of engineering

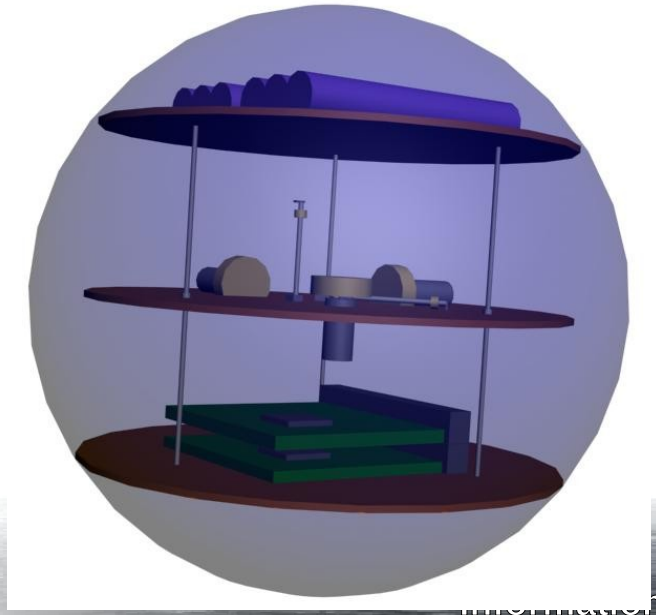
- Concurrent engineering:

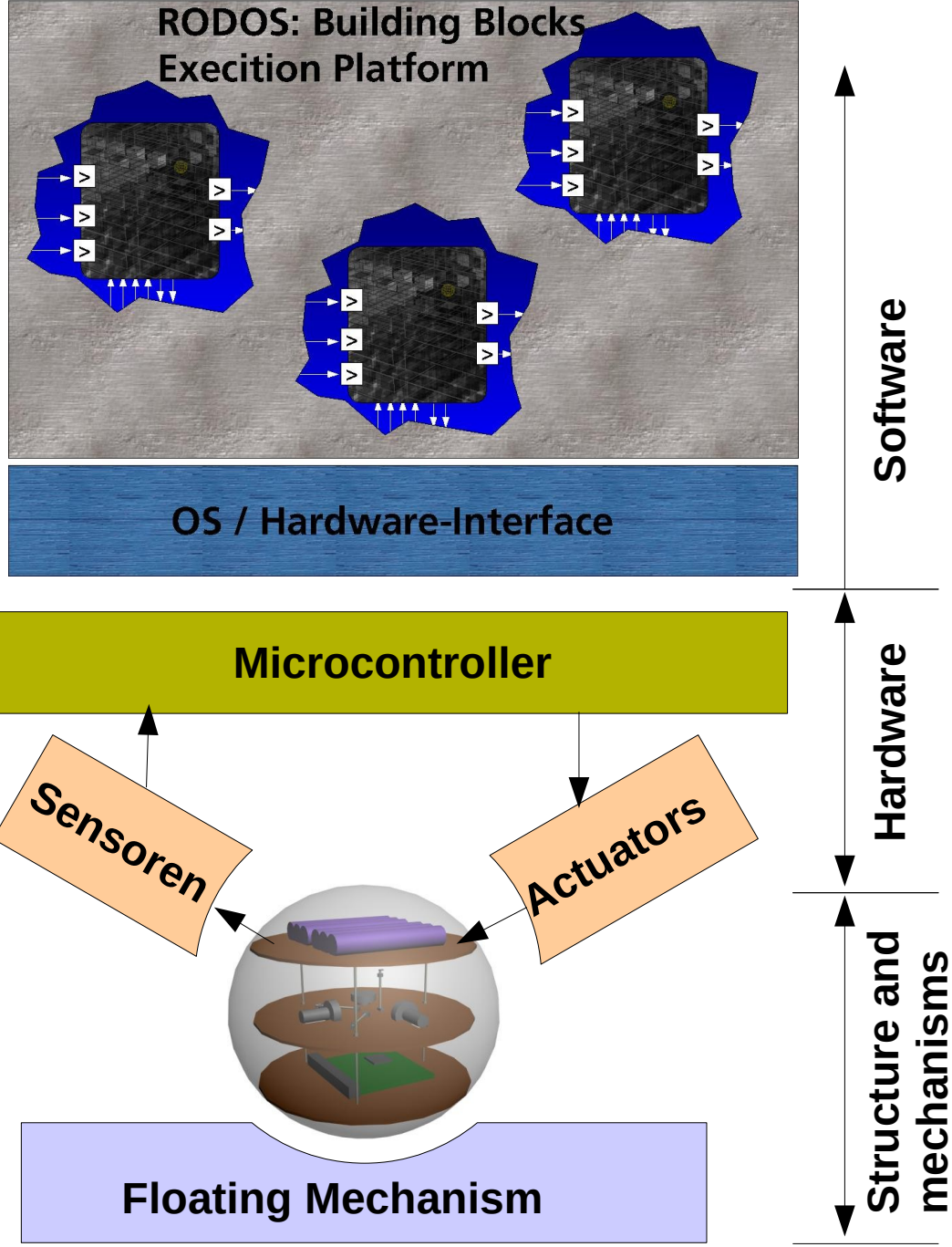



your search easy.

Space Segment





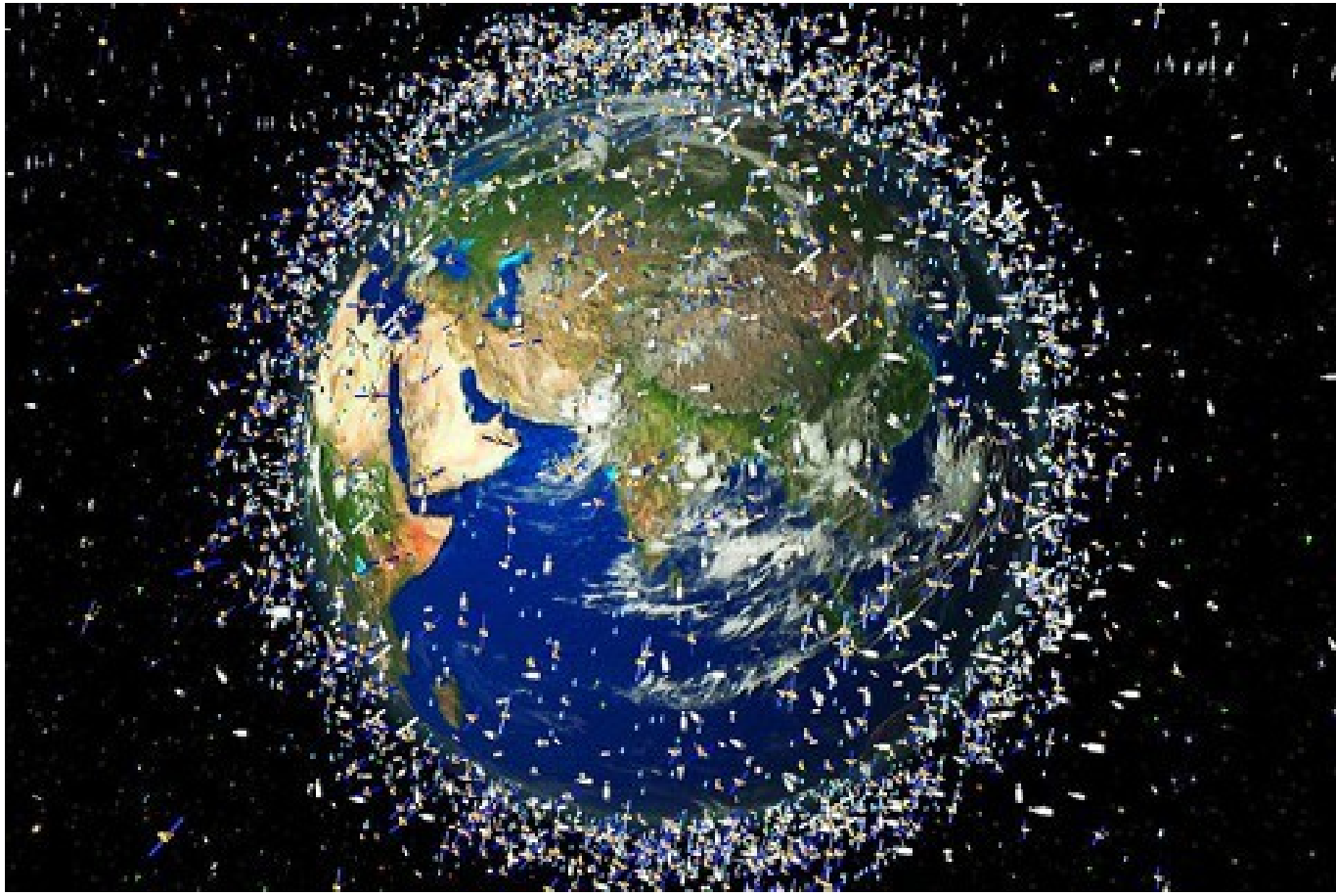

RODOS: Building Blocks
Execition Platform
OS / Hardware-Interface
Microcontroller
Sensoren
Actuators
Floating Mechanism
Software
Hardware
Structure and mechanisms

But…
Back to your task

Build a satellite or a spacecraft

First: What for
What for do we need satellites?

Please say....



What for do we need satellites?

Internet / Telefon → Communication
Television → Kommunikation
GPS & Navigation
Earth science
- Optic
  - Red.. green, blue (RGB)
  - Infrared,
  - UV
  - Multispectral
- Radar
- Lidar

Space Exploration
- Telescopes
- ...

What does the satlite needs to do this functions?

What	**What for**
Kamera	
Antena	
Transmiter	
Reciever	
...	
Sensors?	
Actuators?	

We need things..... and Persons (experts)

Which backgrounds do we need? What is yours?

Mechanics, Electronics, Informatics, Aerospace Technologies, Thermodynamic, Chemistry, Energy, Physics, Mathematics, ….. + Finances + 1x Manager

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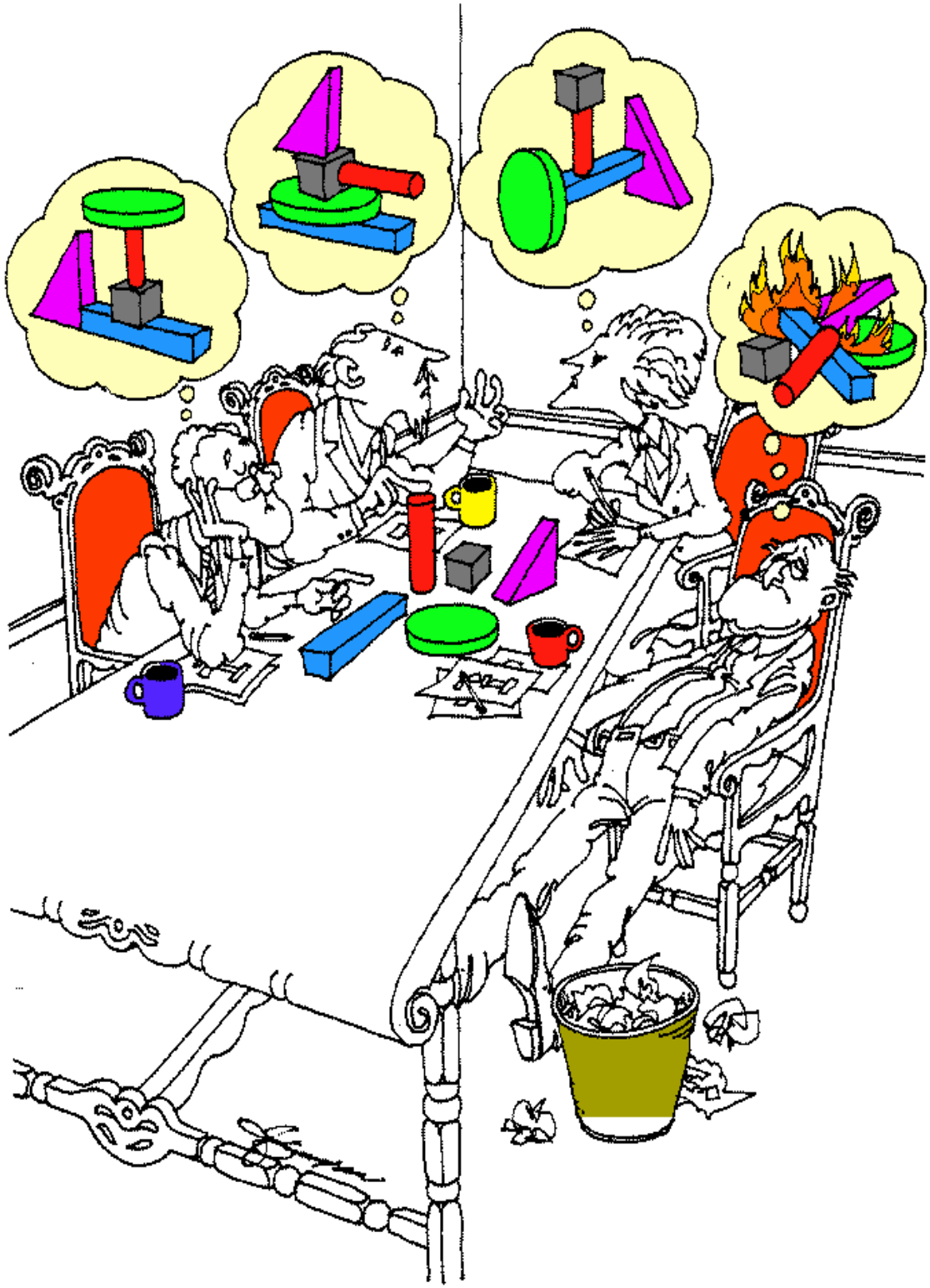

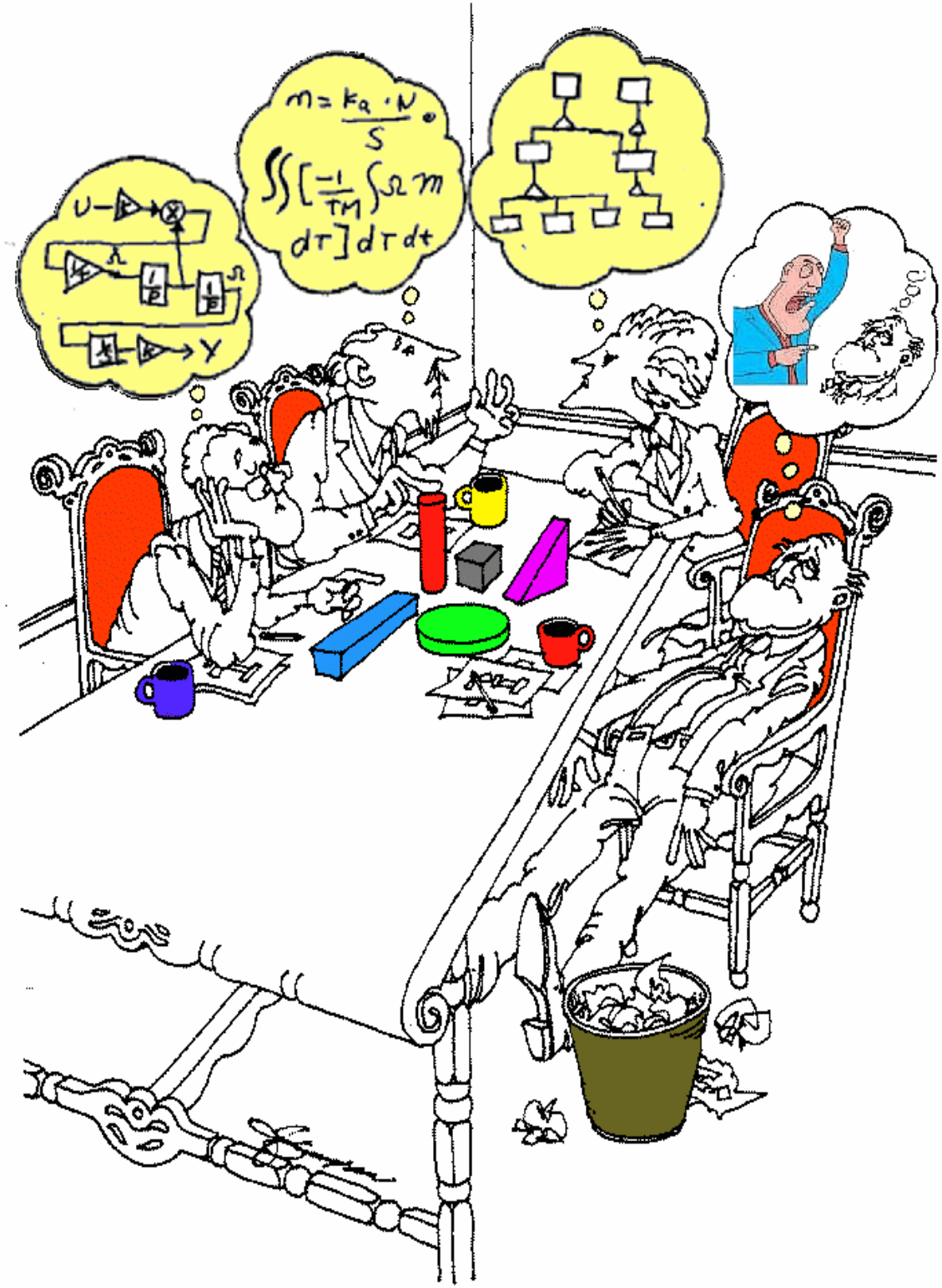

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Interdiciplinary Team

Disciplines:
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we need all! one missing -> will not fly

every one is the most important... me too

Lets build interdiciplinary, International Teams!

Interdiciplinary Team

Disciplines:
- Software
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we need all! one missing -> will not fly

every one is the most important... me too

You are the most important
If you do not do your job
It will not fly!

Danke!

AEROPSACE
INFORMATION
TECHNOLOGY